

DATA ANALYTICS LEVEL III

Day 9: Data Cleaning in Excel — Power Query

Student Activities

How to Use This Document

You are TechHub's analyst. Sales needs the customer list cleaned for a mail-merge campaign — and the export from the old system (customers_raw.csv, download from the LMS) is a mess. Work in Power Query, keep a cleaning log as you go, and NEVER edit the raw file itself.

Activity	What You Submit	Time
Activity 1: Load & Explore	Answers to the 4 exploration questions	30 min
Activity 2: The Cleaning Job	Cleaned workbook + your cleaning log	80 min
Activity 3: Verify, Report & Refresh	Report answers + refresh screenshot	50 min
Learning Journal Entry #8	Your journal reflection	20 min

Activity 1 — Load & Explore (Guided)

Goal: Get the file into Power Query and scout the dirt BEFORE cleaning anything.

1. Data → Get Data → From Text/CSV → customers_raw.csv → TRANSFORM DATA (not Load).
2. Q1: How many rows does the editor report? (Look at the bottom-left corner.)
3. Q2: Click the filter arrow on the city column. List every distinct value you see — exactly as spelled. How many “versions of Manila” are there?
4. Q3: What data type icon does monthly_spend show (ABC or 123)? Why did it import that way? (Look at the values.)
5. Q4: Scroll the email column. Which two customers have a problem you'll need to report?

Submit answers in: Day09_Activity1_[YourName].docx

Activity 2 — The Cleaning Job

Goal: Fix all seven issues IN POWER QUERY, logging each fix.

Do the steps in this order (order matters for #1 vs #2 — think about why):

6. TRIM all text columns (Transform → Format → Trim).
7. REMOVE DUPLICATES on all columns (Home → Remove Rows → Remove Duplicates). Record the row count before and after.
8. CASING: Capitalize Each Word on full_name and city; lowercase on email.
9. VARIANTS: open the city filter dropdown and READ IT FIRST — after step 3, the QC rows now show as “Qc” (Capitalize Each Word title-cased the acronym). Replace Values is case-sensitive, so searching “QC” would silently match nothing. Replace “Qc” → “Quezon City”. Check the dropdown again — how many distinct cities remain?
10. SPLIT full_name by the space delimiter → rename the new columns first_name and last_name.
11. NUMBERS: on monthly_spend, Replace Values , → (nothing), then set the type to Whole Number. Confirm no Error cells appear.
12. TYPES: confirm signup_date is type Date and customer_id is Whole Number.
13. Home → Close & Load.

Your Cleaning Log (required — fill while you work)

#	Issue Found	Power Query Step	Evidence (counts / examples)
1			
2			
3			
4			
5			
6			
7			

Submit: Day09_Cleaned_[YourName].xlsx (workbook with the loaded clean table) + the log (this doc or a sheet)

Activity 3 — Verify, Report & Refresh

Goal: Prove the cleaning worked, answer Sales' questions, and experience the refresh superpower.

Part A — Verification

14. How many unique customers survived cleaning? (Expected: fewer than the raw 28.)
15. Open the city filter on the LOADED table: list the distinct cities. Is every variant gone?
16. Run =MIN and =MAX on monthly_spend. Are the values plausible pesos (not errors, not 0)?

17. Open customers_raw.csv in Notepad. Is it still dirty? What does that prove about Power Query?

Part B — Sales' Questions (use formulas or a quick pivot on the clean table)

18. How many customers per city?
19. Which two customers must Sales contact for missing emails? (IDs and names.)
20. What is the TOTAL monthly spend of the clean list? And the average per customer?
21. Who is the single biggest spender, and how much?
22. Which city spends the most in total — and is it also the city with the most customers? (One sentence on why that's interesting.)

Part C — The Refresh Test

23. A new customer signs up! Open customers_raw.csv in Notepad and add this line at the bottom, then save:

126,Noel Abad,noel.abad@email.com,Pasig,2024-05-15,"2,000"

24. In Excel: Data → Refresh All. Watch the table update. What is the customer count now? Did Pasig appear in the city list — and did YOUR cleaning steps run on the new row automatically?
25. Screenshot the refreshed table for submission.

Submit: Day09_Activity3_[YourName].docx (answers + refresh screenshot)

Learning Journal Entry #8

Goal: Reflect on cleaning as a discipline, not a chore.

1. What does “non-destructive cleaning” mean, and how did today's work prove the raw file was never touched?

2. Why does Remove Duplicates need trimming/standardizing to happen first to be fully effective? Give an example from today's file.

3. You added ONE row to the CSV and clicked Refresh. Describe what happened and why that matters for monthly reports.

4. My confidence with Power Query today (1 = lost, 5 = very confident), and why:

Submit as: Day09_Journal8_[YourName].docx

Submission Checklist

- ☐ **Activity 1:** four exploration answers
- ☐ **Activity 2:** cleaned workbook + completed cleaning log
- ☐ **Activity 3:** verification + report answers + refresh screenshot
- ☐ Learning Journal Entry #8